

K-Medoids Variants: CLARANS

Clustering Large Applications based on RANdomized Search

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Introduction to Clustering

- **Clustering** is an unsupervised data mining technique
 - Groups similar data objects into clusters
 - Objects within the same cluster are **more similar** than those in different clusters

Example:

- Grouping students based on marks
- Grouping customers based on purchasing behavior

What is a Medoid?

- A **medoid** is the most centrally located object in a cluster
- It minimizes the **sum of distances** to other objects in the cluster

Example:

For points {2, 4, 6}

- Mean = 4
- Medoid = **4** (actual data point)

What is CLARANS?

- **CLARANS** stands for

Clustering Large Applications based on RANdomized Search

- A randomized version of K-Medoids
- Combines:
 - PAM's accuracy
 - CLARA's scalability
- Uses **random search** instead of exhaustive search

Key Idea of CLARANS

- Treats clustering as a **graph search problem**
- Each node represents a set of **k medoids**
- Randomly explores neighboring solutions
- Moves to a better solution if found

How CLARANS Works (Steps)

1. Randomly select **k medoids**
2. Assign each object to the nearest medoid
3. Randomly choose a **neighbor solution**
(swap a medoid with a non-medoid)
4. Compute total clustering cost
5. If cost is reduced → move to new solution
6. Repeat until no better neighbor is found
7. Restart with a new random solution (optional)

Example of CLARANS

Dataset:

$\{2, 4, 6, 8, 10\}$, $k = 2$

1. Random medoids $\rightarrow \{4, 8\}$
2. Assign points:
 - Cluster 1: $\{2, 4, 6\}$
 - Cluster 2: $\{8, 10\}$
3. Try swapping medoid 4 with 6
4. If total distance decreases $\rightarrow$ accept new medoids
5. Continue until best medoids found

Advantages of CLARANS

- Works well for **large datasets**
- More **efficient** than PAM
- Less sensitive to **outliers**
- **Better clustering quality than CLARA**

Disadvantages of CLARANS

- Randomized → result may vary
- Still slower than K-Means
- Performance depends on parameter selection

Applications of CLARANS

- Market segmentation
- Image clustering
- Document clustering
- Bioinformatics (gene expression data)
- Customer behavior analysis

Conclusion

- CLARANS is an efficient K-Medoids variant
- Uses **randomized search** to reduce computation
- Suitable for large-scale clustering problems
- Provides a balance between **accuracy and performance**